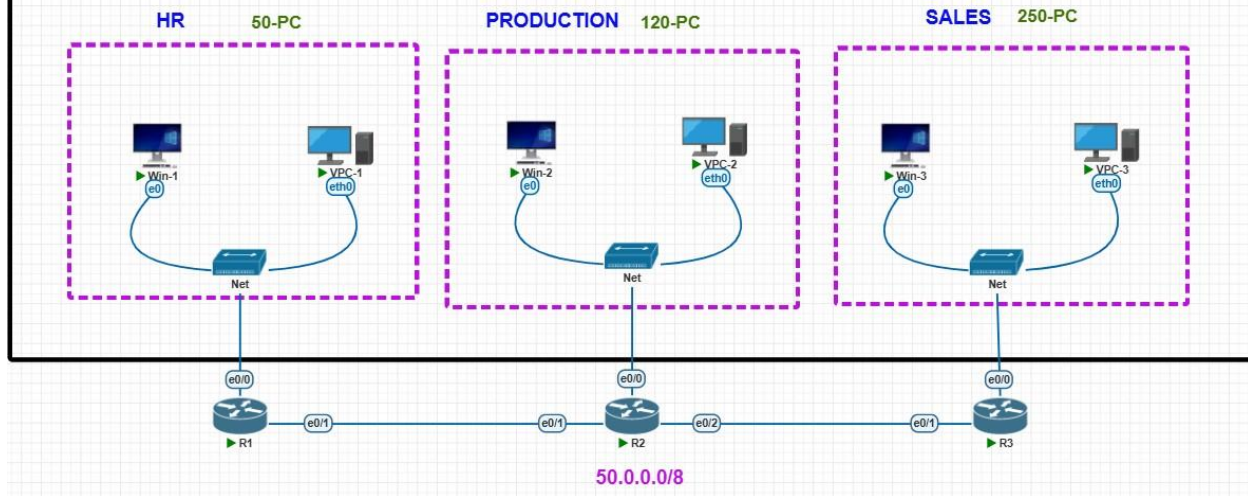


## SUBNETTING



| VLSM Subnetting Scheme — Base Network 50.0.0.0                                 |              |                 |      |            |                 |                |              |
|--------------------------------------------------------------------------------|--------------|-----------------|------|------------|-----------------|----------------|--------------|
| Department/Link                                                                | Hosts Needed | Subnet Mask     | CIDR | Network ID | First Usable IP | Last Usable IP | Broadcast IP |
| SALES                                                                          | 250          | 255.255.255.0   | /24  | 50.0.0.0   | 50.0.0.1        | 50.0.0.254     | 50.0.0.255   |
| PRO                                                                            | 120          | 255.255.255.128 | /25  | 50.0.1.0   | 50.0.1.1        | 50.0.1.126     | 50.0.1.127   |
| HR                                                                             | 60           | 255.255.255.192 | /26  | 50.0.1.128 | 50.0.1.129      | 50.0.1.190     | 50.0.1.191   |
| Router1 - Router2 Link                                                         | 2            | 255.255.255.252 | /30  | 50.0.1.192 | 50.0.1.193      | 50.0.1.194     | 50.0.1.195   |
| Router2 - Router3 Link                                                         | 2            | 255.255.255.252 | /30  | 50.0.1.196 | 50.0.1.197      | 50.0.1.198     | 50.0.1.199   |
| Next Available Network ID: 50.0.1.200 (remaining address space for future use) |              |                 |      |            |                 |                |              |

```

R1
Ethernet0/0      unassigned    YES TFTP  up        up
Ethernet0/1      50.0.1.193    YES manual up        up
Ethernet0/2      unassigned    YES TFTP  down      down
Ethernet0/3      unassigned    YES TFTP  down      down
R1(config)#
R1(config)#
R1(config)#

R2
Ethernet0/0      unassigned    YES TFTP  up        up
Ethernet0/1      50.0.1.194    YES manual up        up
Ethernet0/2      50.0.1.197    YES manual up        up
Ethernet0/3      unassigned    YES TFTP  down      down

R3
R3(config-if)#do sh ip int br
Interface      IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0      unassigned    YES TFTP  up        up
Ethernet0/1      50.0.1.198    YES manual up        up
Ethernet0/2      unassigned    YES TFTP  down      down
Ethernet0/3      unassigned    YES TFTP  down      down
  
```